

# Gender Dynamics in Drone Technology Adoption: Sectoral Differences Across Agriculture, Forestry, and Livestock

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## Abstract

This study examines gender participation patterns in the adoption of drone technology across the agriculture, forestry, and livestock sectors in Spain, Norway, and Greece. We conducted semi-structured interviews with sectoral domain experts involved in drone deployment and identified sectoral differences in gender balance, ranging from relatively inclusive leadership in agriculture to persistently male-dominated practices in forestry and livestock farming. Participants described structural, motivational, and cultural factors as key barriers to women's participation in drone adoption, rather than technological challenges. They also proposed practice-oriented strategies to support more inclusive adoption, including practical demonstrations of drone use on farms, community-based awareness activities through trusted local networks, and training approaches that involve families and acknowledge women's roles in farm decision making. We provide empirically grounded insights into how drone adoption can be made more inclusive through situated training and engagement practices in rural innovation contexts.

## CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design**; **Collaborative and social computing**.

## Keywords

gender and technology, drone adoption, rural production technologies, agriculture, forestry, livestock farming, inclusive design, socio-technical systems

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## 1 Introduction

Drone technologies are increasingly deployed across land-based production sectors such as agriculture, forestry, and livestock farming, supporting activities including crop monitoring, precision spraying, forest assessment, and animal surveillance [9, 11, 16]. As part of broader trends in digital and precision agriculture, drones are promoted as tools that can improve efficiency, reduce physical labour, and enable new forms of data-driven decision-making in rural work practices [2, 3]. In Europe, drone adoption is further supported through policy initiatives and research programmes aimed at accelerating smart and sustainable rural innovation [14].

Despite growing interest in agricultural drones, technology adoption in rural contexts often reflects existing social and organisational structures, including persistent gender inequalities. Although women constitute a substantial share of the global agricultural workforce, their participation in agricultural technology adoption remains comparatively limited [4, 5]. Prior research on drone adoption has largely focused on technical performance, economic benefits, or behavioural intention models [7, 8], while gender is frequently treated as a background variable or omitted altogether [1, 10]. Where gender differences are reported, they are often difficult to interpret due to low female representation or a lack of sector-specific analysis, leaving a limited understanding of how gendered participation varies across agriculture, forestry, and livestock contexts. In addition, prior research has largely examined drone adoption through surveys or farmer-focused studies, while expert perspectives on technology adoption remain underexplored.

To address this gap, we conducted an exploratory qualitative study examining gender participation patterns in drone technology adoption across three rural production sectors (i.e., agriculture, forestry, and livestock) in Spain, Norway, and Greece. Through semi-structured interviews with domain experts actively involved in drone deployment, we investigated how participants described women's participation and perceived barriers within their respective sectors. This cross-sector perspective enables comparison across domains, highlighting how differences in work environments and roles shape participation in drone adoption.

This paper makes three contributions. First, it provides empirical insight into sectoral differences in gender participation in drone adoption across agriculture, forestry, and livestock farming. Second, it shows that barriers to women's participation are perceived as primarily structural, motivational, and cultural rather than technological. Third, it highlights practice-oriented strategies proposed by participants to alleviate gender imbalance in drone adoption.

These include hands-on demonstrations, community-based awareness building, and family-inclusive training approaches that point toward more inclusive pathways for the design and deployment of drone technologies.

## 2 Study Design

We conducted semi-structured interviews with three domain experts representing agriculture, forestry, and livestock farming in Spain, Norway, and Greece, respectively.

### 2.1 Participants

Participants were purposively selected based on their active involvement in drone implementation within their sectors and their role as designated drone pilots in the SPADE project. Recruitment was carried out through the project consortium, and all participants agreed to take part in in-person interviews during scheduled project meetings. Participation was voluntary and uncompensated. Together, the participants brought substantial sectoral and technical expertise relevant to drone adoption:

- **Participant A (Agriculture, Spain, 46)** is a project technician with approximately 12 years of experience in agricultural project coordination and about 3 years of hands-on experience with drone technologies in cooperative farming contexts and a background in environmental sciences.
- **Participant F (Forestry, Norway, 35)** is a research scientist with around 2 years of experience in forestry and 2 years of experience working with drones, supported by a background in mechanical engineering and robotics.
- **Participant L (Livestock, Greece, 65)** is a livestock pilot coordinator with approximately 15 years of experience in livestock farming and about 4 years of involvement in drone technologies for animal monitoring, supported by an engineering background.

### 2.2 Procedure

Interviews lasted 30–45 minutes and addressed: (1) current drone usage and applications, (2) sector-specific benefits and challenges, (3) gender participation patterns and barriers, (4) training and skill development needs, (5) regulatory environment, and (6) future adoption potential. All interviews were audio-recorded with participants' consent and transcribed verbatim for analysis. The interview protocol and data collection procedures were reviewed and approved by the authors' institutional ethics committees.

## 3 Key Findings

### 3.1 General Barriers to Drone Technology Adoption

The interviews identified barriers that affect the adoption of drone technology across all sectors.

**3.1.1 Skills and Motivation Barriers.** Participants consistently indicated that barriers to drone adoption were not related to inherent technical ability, but rather to digital literacy gaps, motivation, and awareness. Participant L stated that *“there are no inherent skill barriers.”* Instead, the interviews revealed digital literacy gaps as a key challenge. Participant A highlighted the situation: *“The farmers in*

*Mallorca are older people. Maybe some of them don't use smartphones very well. And they are aware of that. They want to use tools. But the main problem is the digital gap.”* Together, these accounts indicate that motivation and awareness, rather than technical ability, shape early engagement with drone technologies.

**3.1.2 Regulatory Barriers.** Regulatory complexity emerged as a significant obstacle to drone adoption across sectors. In particular, compliance requirements and permitting procedures were perceived as time-consuming and difficult to navigate, especially for small-scale operations. Participant A noted that in Spain, *“maybe the regulation. Because in Spain, for example, for some of the farms we have to take five permissions,”* illustrating how regulatory complexity can discourage adoption even when technical interest and readiness are present.

**3.1.3 Institutional Support Gaps.** Participants highlighted a lack of sector-specific institutional support for drone technology adoption. While national and regional initiatives promoting digitalization exist, these were described as largely generic and insufficiently tailored to the operational realities of specific sectors. As a result, participants perceived limited guidance and continuity of support beyond pilot projects or demonstration activities, constraining sustained adoption, particularly at the operational level.

**3.1.4 Technology Maturity Barriers.** Differences in the maturity of drone technologies across sectors were also identified as affecting adoption. Participant L noted that livestock farming, in particular, *“does not yet have mature, widely adopted drone solutions,”* suggesting that the limited availability of sector-specific technologies constrains broader uptake in this domain.

### 3.2 Gender Effects of Adoption Barriers

Our analysis reveals clear differences in gender participation across sectors.

**Agriculture (Spain).** A split pattern was observed. Women were strongly represented in leadership roles in the Balearic Islands. Participant A explained: *“The presidents of the agri-food cooperatives are women, but the workers here are men.”* Women were also described as working as agronomists and participating in drone training programs: *“They go to the course to know how it works... the technology... but then the ones that go to the field to fly the drones are men.”*

**Forestry (Norway).** Forestry showed consistent male dominance across roles. Participant F stated, *“It is fairly male-dominant,”* noting that this pattern extends beyond traditional forestry work: *“Drone technology, AI, and engineering are currently even more male-dominated fields, largely due to societal norms.”*

**Livestock (Greece).** Livestock farming exhibited the most pronounced gender imbalance. Participant L described that *“women working in livestock farming are relatively few,”* with an even wider gap in technology use: *“Women using technology in livestock are even fewer.”*

Together, these sectoral accounts highlight that women's participation varies not only in degree, but also in the types of roles through which engagement with drone technologies occurs.

While many of the barriers identified above affect drone adoption broadly, our findings indicate that these constraints are not experienced uniformly. Rather, they tend to have gendered effects,

disproportionately limiting women's participation due to existing divisions of labour, differential access to resources, and socially embedded role expectations. In the following subsections, we examine how these sectoral patterns are shaped by the interaction between everyday work contexts, access to learning opportunities, and institutional conditions, and how these factors contribute to persistent gender imbalances in drone use.

**3.2.1 Gendered Effects of Physical Demands and Work Context.** While harsh working conditions were described as a general barrier to drone adoption, participants indicated that these conditions have gendered effects in practice, particularly discouraging women's participation in livestock farming. Participant L explained that limited participation is *"not due to technical barriers but rather motivation, education level, and the harsh working environment."*

At the same time, drone technologies were framed as a potential enabler of more inclusive participation, particularly when they reduce the need for constant physical presence in demanding work environments. Participant L suggested that *"technology could be a factor that could alleviate access to a more balanced gender access in technology and livestock, because technology is user-friendlier than the traditional way."* In particular, remote monitoring was described as making work contexts more accessible, potentially supporting improved work–life balance and family inclusion.

**3.2.2 Gender Effects of Skills, Motivation, and Learning Opportunities.** While digital literacy and motivation were identified as general barriers to drone adoption, participants suggested that these factors have gendered effects in practice. Women were frequently described as participating in training and certification activities but having fewer opportunities for hands-on operation and informal learning. As Participant A explained, *"They go to the course to know how it works... the technology... but then the ones that go to the field to fly the drones are men."*

As a result, women's engagement with drone technologies was more often positioned in advisory, educational, or administrative roles rather than operational use. Participants noted that even when women possess formal qualifications, limited access to equipment and field-based experience constrains their ability to translate training into everyday practice. This uneven access to practical engagement shapes motivation and perceived relevance of drone technologies, reinforcing existing role divisions rather than enabling more balanced participation.

**3.2.3 Institutional and Regulatory Constraints.** Institutional and regulatory arrangements were described as shaping gendered access to drone technologies in practice. Although these frameworks apply broadly across sectors, their effects were perceived as particularly restrictive for women in contexts where they are already underrepresented, such as forestry and livestock farming. In these settings, generic support initiatives often fail to account for differences in access to training, equipment, and operational opportunities. As a result, institutional and regulatory frameworks may unintentionally reinforce established divisions of labour, limiting women's opportunities to move from supportive or peripheral roles into operational engagement with drone technologies.

**3.2.4 Family-Centered Inclusion Approaches.** Family-centered and community-based approaches were described as important for supporting more inclusive technology adoption in rural contexts, where farming and livestock practices are embedded in shared household responsibilities. Rather than framing adoption as an individual decision, participants emphasized the role of families in shaping technology engagement. Participant L suggested that adoption strategies should *"start bottom-up by understanding barriers in livestock farming, then create awareness sessions involving families, women, and children."* Such approaches were seen as broadening participation by aligning technology adoption with everyday family dynamics and collective decision-making.

### 3.3 Training and Awareness Needs

Participants emphasized that effective drone adoption depends not only on access to technology, but also on how awareness and training activities are structured and embedded in local contexts. Across sectors, several recurring approaches were described.

First, participants stressed the importance of building awareness of relevance and potential benefits before introducing formal technical instruction. As Participant L noted, *"training should start with awareness-building rather than formal instruction,"* emphasizing that users need to understand what the technology can do before learning how to operate it.

Second, direct and hands-on interaction with drone technologies was described as essential for building confidence and practical understanding. Participant F highlighted that *"hands-on workshops are very effective,"* particularly for reducing uncertainty and increasing perceived accessibility.

Third, locally grounded and community-based engagement was described as more effective than centralized seminars, especially in rural contexts. Community meetings and on-site demonstrations were seen as better aligned with everyday practices and more accessible for older generations.

Participants also highlighted the value of combining multiple learning formats. Physical meetings and printed materials were described as important entry points, while digital resources such as videos and online tutorials were seen as useful for supporting gradual transitions from analog to digital practices. This combination was viewed as enabling sustained learning without overwhelming new users.

Finally, inclusive awareness and training activities involving families were described as particularly important in farming and livestock contexts. Participant L emphasized the value of *"creating awareness sessions involving families, women, and children,"* reflecting the role of shared decision-making and household dynamics in shaping technology adoption.

#### Key Implications for Design:

- (1) User-friendly interfaces can support confidence-building and lower perceived entry barriers.
- (2) Community-based and family-inclusive training approaches can be more effective than individual, skills-focused programs.
- (3) Leveraging existing, more gender-balanced professional networks (e.g., veterinary services or agricultural cooperatives) can support broader and more inclusive adoption pathways.

## 4 Discussion

Our findings show that gender participation in drone adoption differs across the agriculture, forestry, and livestock sectors, and that these differences are shaped primarily by social, institutional, and contextual factors rather than by technological constraints alone. Studies from Germany [7], Hungary [1], and China [15], for example, report gender differences in adoption patterns while noting that these differences cannot be readily attributed to technical capability or perceived ease of use. Our findings extend this work by demonstrating how such gendered differences manifest unevenly across rural sectors, even within European contexts.

Beyond sectoral differences in adoption patterns, our findings indicate that male dominance persists across all three domains, although it takes different forms across regions and professional contexts. In the agricultural setting of the Balearic Islands, the presence of women in administrative, advisory, and agronomist roles suggests that engagement with drone technologies is shaped by occupational positioning rather than by access to training alone. Similar role-based patterns have been identified in earlier studies, where women are more likely to participate in coordination and knowledge-related activities while remaining underrepresented in operational technology use [5, 10].

The forestry and livestock sectors further illustrate how work environments and sector-specific practices shape participation in technology adoption. In forestry, the persistent male dominance reported by participants reflects broader patterns observed in European smart agriculture and forestry initiatives, where digital innovations often reproduce existing workforce structures when gender considerations are not explicitly addressed [14]. In livestock farming, gender imbalance appears particularly pronounced, shaped by physically demanding conditions, long working hours, and limited institutional support. These findings are consistent with research showing that technology-intensive livestock systems can reinforce existing inequalities in the absence of gender-responsive extension services and innovation programmes [6, 12].

Across sectors, barriers to participation were not framed in terms of technical skill or system usability. Instead, motivation, awareness, confidence, and perceived relevance of drone technologies emerged as more influential factors. This reframing aligns with adoption research that emphasizes the role of social context and locally meaningful use cases in shaping technology uptake [13, 16]. From this perspective, uneven participation reflects how technologies are introduced, supported, and embedded in everyday practices rather than a lack of individual capability.

Finally, the training and awareness strategies described by participants point toward bottom-up approaches that situate drone adoption within family and community contexts. Community-based demonstrations, on-farm engagement, and visual examples were consistently described as more effective than formal classroom-based training. This emphasis on experiential and socially embedded learning echoes earlier findings that farmers are more likely to engage with drone technologies when they can observe concrete benefits in familiar settings [13, 16].

## 4.1 Limitations and Future Work

This study provides exploratory insights into gender participation in drone adoption across rural production sectors based on interviews with domain experts directly involved in drone deployment. While this perspective offers valuable insight into implementation, coordination, and training practices, it does not capture the full range of experiences of farmers or other end users, and the findings should therefore be interpreted at the level of sectoral and organisational dynamics. Future work could extend this research through larger-scale qualitative or quantitative studies that include a broader range of stakeholders. In particular, the practice-oriented strategies identified by participants provide a basis for designing and empirically evaluating targeted interventions across different sectoral and cultural contexts.

## 5 Conclusion

This study demonstrates that gender participation in drone adoption differs across agriculture, forestry, and livestock sectors, reflecting sector-specific work practices, organisational structures, and social contexts rather than technological constraints alone. By drawing on the perspectives of domain experts involved in drone deployment, the findings show how adoption pathways are shaped by everyday practices, institutional conditions, and role divisions within rural production systems. By highlighting training and engagement approaches proposed by participants, this work points toward concrete pathways for supporting more inclusive drone adoption and suggests that efforts to promote digital innovation in rural contexts should attend not only to technological development, but also to the social and organisational conditions through which technologies are introduced and embedded in practice.

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